SQL CASE STATEMENTS

EXERCISE 3 FOCUS CASE. Classification. Logic

01 The products Table

Table products

product_id	product_name	price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

'Expensive' if price > 1000
'Mid-range' if price between 100 and 1000
'Budget' if price < 100

(1) SELECT product_name, price

```
CASE
    WHEN price < 100 THEN 'budget'
    WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
    WHEN price > 1000 THEN 'Expensive'

    END AS price_category

FROM products;
```

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

(2) The orders Table

Orders

order_id	customer_name	amount
1	Alice	150.00
2	Bob	500.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

'Highvalue' for orders ≥ 1000
'Mediumvalue' for orders 500–999
'Lowvalue' for < 500

(2) SELECT customer_name, amount

CASE

WHEN amount < 500 THEN 'Low value'

WHEN amount BETWEEN 500 AND 999.99 THEN ~~'Medium range'~~ 'Medium value'

WHEN amount ≥ 1000 THEN 'High value'

END AS order_value_category

FROM orders;

Results table.

Customer_name	amount	Order_value_category
Alice	150.00	Low value
Bob	500.00	~~Low~~ Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200.00	High value

The employees Table

Table: Employees

(3)

emp_id	emp_name	department	Salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

If 'IT' & salary > 80000 → 'Senior IT'

If 'HR' & salary > 55000 → 'Experienced HR'

Otherwise 'Staff'

SELECT emp_name, department,

Salary

CASE WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'

WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'

ELSE 'staff'

END AS position_level

FROM employees;.

Results Table for num3

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Experienced HR

The students Table Table : students

student_id	student_name	Score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

≥90 -> 'A'
80-89 -> 'B'
70-79 -> 'C'
60-69 -> 'D'
<60 -> 'F'

④ SELECT student_name, score

```
CASE
    WHEN score ≥ 90 THEN 'A'
    WHEN score BETWEEN 80 AND 89 THEN 'B'
    WHEN score BETWEEN 70 AND 79 THEN 'C'
    WHEN score BETWEEN 60 AND 69 THEN 'D'
    WHEN score < 60 THEN 'F'
END AS grade
FROM students;
```

⇓

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

The deliveries Table

⑤

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

- ≤ 30 mins → 'Fast'
- 31-60 mins → 'On time'
- > 60 mins → 'Late'

⑤
```
SELECT delivery_id, delivery_time_minutes
    CASE
        WHEN delivery_time_minutes ≤ 30 THEN 'Fast'
        WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On time'
        WHEN delivery_time_minutes > 60 THEN 'Late'
    END AS performance;
FROM deliveries;
```

⇓

delivery_id	delivery_time_minutes	performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

⑥ The tickets Table

ticket_table	issue_type	priority
1	Login	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

3 → 'High'
2 → 'Medium'
1 → 'Low'

⑥ SELECT issue_type, priority

```
CASE
    WHEN issue_type = 3 THEN 'High'
    WHEN issue_type = 2 THEN 'Medium'
    WHEN issue_type = 1 THEN 'Low'
END AS priority_label
FROM tickets;
```

Results

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Gmail error	2	Medium
Password reset	1	Low

⑦ The attendance Table

student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

- ≥ 90% – 'Excellent'
- 75-80% – 'Good'
- < 75% – 'Needs improvement'

⑦ SELECT student_id,

```
(days_present/total_days *)100 As attendance_percentage,
CASE
    WHEN(days_present/total_days) *100 >= 90 THEN 'Excellent'
    WHEN (days_present/total_days) *100 >= 75 THEN 'Good'
    WHEN days=present/total_days) *100 < 75 THEN 'Needs Improvement'
END AS attendance_status
FROM attendance;
```

Results_table

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs improvement
3	96	Excellent
4	50	Needs improvement
5	100	Excellent

(8) The products_inventory Table

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

0 -> 'Out of stock'
1-5 -> 'Low stock'
>5 -> 'In stock'

(8)
```
SELECT product_id, stock_qty
    CASE
        WHEN stock_qty = 0 THEN 'Out of Stock'
        WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
        WHEN stock_qty > 5 THEN 'InStock'
    END AS stock_status
FROM products_inventory ;
```

Results_table

product_id	stock_qty	stock_status
1	5	Instock
2	0	Out of stock
3	125	Instock
4	10	Instock
5	3	Low stock

9 The class Table

class_id	subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

≥ 25 → 'Large'
10–24 → 'Medium'
< 10 → 'Small'

9
```
SELECT subject, enrolled_students
    CASE
        WHEN enrolled_students < 10 THEN 'Small'
        WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
        WHEN enrolled_students >= 25 THEN 'Large'
    END AS class_size_category
FROM class;
```

⇓

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10 The payments table

payment_id	amount	payment_method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	paypal
5	300.00	Cash

- If payment_method = 'cash' & amount ≥ 200 → 'Eligible for discount'
- Otherwise → 'Not eligible'

(10)
```
SELECT payment_id, amount,
       payment_method
    CASE
        WHEN payment_method = 'cash' AND amount ≥ 200 THEN
        'Eligible for Discount'
        ELSE 'Not eligible'
        END AS discount_eligibility
FROM payments;
```



payment_id	amount	payment_method	discount_eligibility
1	50.00	Card	Not eligible
2	200.00	Cash	Eligible for discount
3	150.00	Card	Not eligible
4	75.00	Paypal	Not eligable
5	300.00	Cash	Eligible for discount